「Logistic equation.

Find the explicit solution $y = y(t) \in C^1(0,\infty)$ satisfying

$$\begin{cases} y' = y - y^2, & t > 0 \\ y(0) = y_0 \in \mathbb{R}. \end{cases}$$

Sol). If either $y_0 = 0$ or $y_0 = 1$, then $y \equiv 0$ and $y \equiv 1$ are solution for the given initial condition, respectively. And, by the uniqueness of the solution of ODE, the other solutions $y(t)$ cannot intersect $y \equiv 0$ and $y \equiv 1$.

It provides there are three case.

1). $y_0 > 1$

observe that

$$y' = y - y^2 \Rightarrow \frac{y'}{y} + \frac{y'}{1-y} = 1 \Rightarrow \int_0^t \frac{y'}{y} - \frac{y'}{y-1}\, dx = \ln|y| - \ln|y-1| \Big|_0^t$$

$$= \ln\frac{y(t)}{y(t)-1} - \ln\frac{y_0}{y_0-1} = t.$$

Take $C = \ln\frac{y_0}{y_0-1}$ (>0) Then

$$\ln\frac{y(t)}{y(t)-1} = t + C \Rightarrow 1 - \frac{1}{y(t)} = e^{-t-C} \Rightarrow \frac{1}{1-e^{-t-C}} = y(t)$$

2). $0 < y_0 < 1$

Similarly, but concern the sign,

$$\int_0^t \frac{y'}{y} - \frac{-y'}{1-y}\, dx = \ln|y| - \ln|1-y| \Big|_0^t = \ln\frac{y(t)}{1-y(t)} - \ln\frac{y_0}{1-y_0} = t.$$

Take $C = \ln\frac{y_0}{1-y_0}$. Then

$$\ln\frac{y(t)}{1-y(t)} = t + C \Rightarrow \frac{1}{y(t)} - 1 = e^{-t-C} = y(t) = \frac{1}{1+e^{-t-C}}.$$